

Project Executable Files

Date	1 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – Crop Recommendation System
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA (No database used)
5	Environment configuration file (.env.example or similar)	No (Not required)
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	No
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

OptiCrop/

```

|
|— app.py
|— crop_model.pkl
|— requirements.txt
|— README.md
|
|— dataset/
|   |— notebook/
|       |— Crop_recommendation.csv
|       |— crop_prediction.ipynb
|       |— train_and_save.py
|       |— flask_app.py
|       |— templates/
|           |— index.html
|       |— static/
|           |— css/
|           |— images/
|           |— js/
|
|— .gitignore
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://opticrop-wii6.onrender.com/
Login Credentials (Demo)	Not Required (Public Access)
Platform / Hosting Provider	Render

Repository Link	https://github.com/Bhavagna2007/OptiCrop
Demo Video Link	

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

1. Clone the GitHub repository.
2. Open the project in Visual Studio Code.
3. Create and activate a Python virtual environment.
4. Install the required dependencies using:

```
pip install -r requirements.txt
```

5. Start the Flask application:

```
python app.py
```

6. Open the application in your web browser:

```
http://127.0.0.1:5000/
```

7. Enter the soil parameters:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Temperature
- Humidity
- pH
- Rainfall

8. Click **Predict Crop** to view the recommended crop and its description.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Predictions are limited to crops available in the training dataset.	Can be improved by training with a larger dataset.

2	The application requires manual entry of soil parameters.	Future versions can integrate soil sensors or IoT devices.
3	The current version does not provide fertilizer or irrigation recommendations.	Planned as a future enhancement.